

Model Research & Selection

Introduction

Selecting the right AI model is one of the most important decisions in developing any Natural Language Processing (NLP) application. The performance of the application depends heavily on the models used, as they directly affect response quality, processing speed, and overall user experience. A model that is highly accurate but too slow may not be suitable for a real-time application, while a very fast model with poor accuracy can produce unreliable results.

For the Personalized Networking Assistant, we needed two different NLP capabilities. The first was to understand the content of an event description and identify its theme. The second was to generate professional and engaging conversation starters that users can use while networking with new people. Since these two tasks have different objectives, we evaluated separate models for each before making our final selection.

Theme Extraction: Why DistilBERT Was Chosen

For identifying the theme of networking events, we selected DistilBERT. During our research, we considered several transformer-based models available through the Hugging Face library. Our main requirement was a model that could accurately classify event descriptions into meaningful categories without requiring us to create and train a custom dataset.

One of the biggest advantages of DistilBERT is its support for zero-shot classification. This allows the model to classify text into user-defined categories even if it has never been specifically trained for those categories. Since networking events can cover a wide range of topics such as technology, business, healthcare, education, finance, entrepreneurship, and many others, this flexibility is extremely valuable.

Another reason for selecting DistilBERT is its balance between accuracy and speed. Compared to larger transformer models like BERT, DistilBERT is significantly smaller while maintaining most of the original model's performance. This means it can process event descriptions much faster, reducing response time for users without sacrificing much accuracy.

Resource efficiency was another important factor in our decision. Because DistilBERT requires less memory and computational power, it is easier to deploy on standard hardware and cloud environments. This makes it a practical choice for a web application where quick responses and efficient resource usage are essential.

Conversation Generation: Why GPT-2 Small Was Chosen

For generating conversation starters, we selected GPT-2 Small. The objective of this component is not to generate long paragraphs but to create short, natural, and context-aware opening

messages that help users start professional conversations at networking events.

We evaluated different text generation models based on factors such as text quality, computational requirements, ease of deployment, and compatibility with our application. While larger language models can produce more sophisticated text, they also require significantly more computing resources and longer processing times.

GPT-2 Small offered the best balance between quality and efficiency. It generates coherent and contextually relevant conversation starters while remaining lightweight enough to run on standard CPU hardware. This makes it suitable for applications where GPU resources may not always be available.

Another advantage of GPT-2 Small is its seamless integration with the Hugging Face Transformers library. The model can be loaded and used with only a few lines of code, making development and deployment straightforward.

Why These Models Were Selected

After comparing multiple options, we selected DistilBERT and GPT-2 Small because together they provide the best balance between performance, speed, and deployment efficiency.

DistilBERT enables fast and accurate event theme classification without requiring additional training data, making it highly adaptable to different types of networking events. GPT-2 Small complements this by generating meaningful conversation starters quickly while keeping computational costs low.

Using lightweight models also improves scalability. As the number of users grows, the application can continue delivering quick responses without requiring expensive hardware upgrades.

Conclusion

After evaluating different NLP models, we concluded that DistilBERT is the most suitable choice for event theme extraction because of its zero-shot classification capability, fast inference speed, lightweight architecture, and reliable accuracy. Similarly, GPT-2 Small is well suited for generating conversation starters because it produces natural text while remaining computationally efficient and easy to deploy.

Together, these models meet the functional requirements of the Personalized Networking Assistant by providing accurate event understanding and helpful conversation suggestions.